Text 1

For thousands of years, donkeys have been critical for propelling human civilizations forward. They've helped pull wheeled vehicles, carry travelers and move goods across the world. But where and when these animals first became intertwined with humans has been a mystery. Now, researchers have used genomes of over 200 donkeys to trace their domestication back to a single event around 7 000 years ago in East Africa—about 3 000 years before humans tamed horses. The team published their findings in the journal Science this month.

"Through their DNA, the animals are telling their history themselves," co-author Samantha Brooks, an equine researcher at the University of Florida, says in a statement. "We usually only get the human's side of history through written accounts, but of course written history does not always record exactly how something happened. Looking at these DNA sequences, we get a biological testimony to the environment these animals lived in and the experiences they survived."

The researchers examined 207 genomes from modern donkeys living in 31 countries across the globe. They also looked at genomes from 15 wild equids and 31 earlier donkeys that lived between about 4 000 and 100 years ago. The team reconstructed the animals' evolutionary tree and used computer models to pinpoint the domestication event: when herders in Kenya and the Horn of Africa tamed wild asses. They then traced how the animals spread across the rest of the continent into Europe and Asia about 2 500 years later.

Though it's still unclear why the original domestication happened, Science News' Freda Kreier reports that the event coincided with the Sahara growing larger and drier. "Donkeys are champions when it comes to carrying stuff and are good at going through deserts," co-author Ludovic Orlando, an evolutionary biologist at Paul Sabatier University in France, tells the publication. Prehistoric humans may have enlisted donkey's help in navigating the expanding Sahara.

Researchers say these findings could help put donkeys in the spotlight. The animals could benefit from more research; Currently, there are no published genomes from donkeys located south of the equator in Africa. But understanding where the animals were first domesticated could guide archaeologists to a narrower region to search for insights about the original tamed donkeys.

Not only does understanding the equines' genetic makeup help reveal their contribution to human history, but it also might improve their management in the future, as climate change alters the planet's environment, write the authors.

Text 2

There's no business like show business—but in Los Angeles, it feels like there's no business at all.

If that sounds melodramatic, consider this; The Art Directors Guild, a labor union representing about 3 000 film workers, has suspended a training program and issued a statement explaining that "we cannot in good conscience encourage you to pursue our profession while so many of our members remain unemployed." The guild's pessimism is a reaction to Hollywood's decline, which is reaching a critical point for the industry and Southern California.

Production has been slipping away from Hollywood since the 1950s, but the effects have never been more apparent than at present. Other regions in the United States, Canada and Europe have steadily increased incentives to attract TV shows and movies, leaving California in the dust. Georgia offers up to 30% in transferable tax credits on film and TV production costs, plus an additional 10% increase on the base tax credit if the project includes a Georgia promotional logo.

Even as California lost a huge volume of production to other locations, there was still plenty of film production taking place in Los Angeles before this year. We were kept afloat by "peak TV": the glut of content that was required by the explosion of streaming services.

But 2022 was the peak of peak TV. Back then platforms such as Netflix, Amazon and Apple TV hemorrhaged billions of dollars to generate content to attract new subscribers, resulting in 633 scripted series being released that year. As the streamers' emphasis changed from subscriber growth to profitability, prices for the services went up and the number of new shows went down to 481 released in 2023, with the number expected to dip into the 300s within a few years.

If productions in Southern California dip below a critical level for too long, the industry's essential talent will drift away along with enormous sums of revenue. Persuading studios to film here would become much more challenging if we couldn't offer a deep bench of local film workers, on-screen talent and local businesses that support the entertainment industry.

That's why the California Film Commission and its Los Angeles counterpart, Film LA, should act now, before it's too late. Those agencies and other government bodies should dramatically improve incentives to keep our current shows and attract new productions to Los Angeles. Let's go on with the show... and make sure the show doesn't go on without us.

Text 3

The pioneers of wireless saw it as a gift to all the people. Sir John Reith said that it would end "isolation of the spirit" and rejoiced: "It does not matter how many thousands may be listening, there is always enough for others... the genius and the fool, the wealthy and the poor listen simultaneously."

Between two great wars this technological innovation built a new kind of national consciousness. Opening this week, a book and exhibition curated by Beaty Rubens at the Bodleian in Oxford records how radio changed everyday life from 1922 to 1939. She draws on letters, diaries and fiction, and a 1939 field notebook of verbatim audience research by Winifred Gill.

There's fun in testimonies of people enjoying the sheer newness. A cartoon mocks a group failing to converse because they're all in headphones. People report that broadcast music made workmen whistle new tunes. A woman says there have been fewer street fights since the arrival of the wireless but also less stopping and "talking on the brush handle".

By and large the wireless was welcome. I loved the man from the Thirties research who found that wireless suddenly offered "a lot of variety... things I thought I'd never be interested in... ice hockey, perhaps". True: for more than 80 pre-digital years, linear speech broadcasting brought the gift of serendipity, random enlivening of a car journey or dull manual task. In my own book about radio I recorded how, on one drive: "I caught up with the news, learnt some 17th-century history, and was startlingly educated by an unpretentious programme on the history of the stethoscope."

But radio's enriching serendipity is ebbing. With multiple networks and countless podcasts, a smart-phone user selects what to hear and when. And while it is wonderful to take a walk with anything in your headphones, infinite choice encourages us to shrink into niche interests and sympathetic beliefs.

A hundred years on from Marconi and Reith, is the art of mere listening endangered? Some will say the audiobook boom revives it, though I suppose you can then worry about the decline of reading. But inventions shape all of us and it is worth noticing when techno-social habits do change, and asking whether to control them a bit or shield the youngest. Whatever we do, innovation will happen. Today we fret about the isolating culture of smartphone-staring and selfie-vanity, but already in 1939 there was that lady regretting how, when all her street got wirelesses, it lost the neighbourly habit of "talking on the brush handle". It's enough to make a person put down the smartphone and go out front with a yard broom.

Text 4

When Tom Swetnam joined the U.S. Forest Service in the 1970s, his mandate was to "put everything out," he recalled. But when Swetnam enrolled in graduate school at the University of Arizona's Laboratory of Tree-Ring Research, he was surprised to find a record of repeated blazes dating back hundreds of years before European colonists arrived on the continent. Some of the trees he analyzed bore more than 20 fire scars among their rings.

The fact that fires happened so often meant they couldn't have been severe enough to kill most trees. Instead, a growing body of research showed that frequent, low-severity fires made many ecosystems healthier. They rid the forest of dead and sick trees, reducing competition and curbing the spread of disease. Because flammable material couldn't build up on the landscape, blazes tended to move slowly and peter out when they reached the footprints of previous burns.

In 2022, Swetnam and other scientists teamed up to compile a database of fire-scarred trees from across the continent. Their North American tree-ring fire-scar network (NAFSN) provided the basis for a study published last month. In the study, the researchers compared the historical fire cadence with the wildfires recorded over the past few decades, and uncovered a striking shortfall. The NAFSN sites experienced less than a quarter of the number of fires that would have been expected without fire suppression.

This deficit is a testament to the effectiveness of modern firefighting, said Kelly Martin, a past president of the International Association of Wildland Fire. Yet the combined consequences of suppression and climate change have eroded humanity's ability to suppress fires, particularly those that ignite under the most dangerous weather conditions. To prevent entire ecosystems from going up in smoke, Martin said, people must bring healthy fire back to places that need it. At Yosemite National Park, Martin oversaw the use of what are known as prescribed burns to make the landscape more resilient. These fires were carefully planned and intentionally ignited during periods when weather kept the blazes easy to control, and helped eliminate some of the fuel that had built up around important park facilities. Research shows that these prescribed burns make subsequent wildfires less severe, even if later fires happen under the most dangerous weather conditions.

Yet even as scientists and public officials increasingly agree on the need for more fires in our forests, climate change is making this tactic more challenging, experts said. "It's a double-edged sword because wildfires are getting more severe and larger under climate change and we need this work even more, but then the work gets more challenging," said Susan Prichard, a fire ecologist at the University of Washington.